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Department of CSE

B.Tech. CSE Trimester -5

2520030378

MUVVA LOHITHA

Cloud Infrastructure and Services

25CS2206E

LAB RECORD

2026

Experiment -2

AIM:2A: Enabling Multi-Factor Authentication to Secure Your Access

Description:

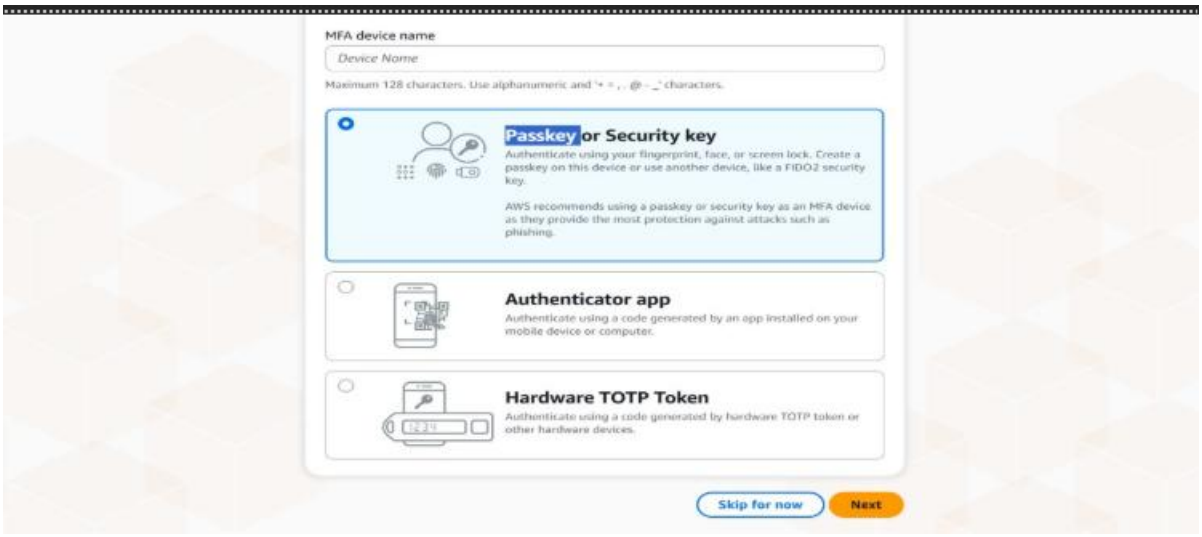
Multi-Factor Authentication (MFA) is a security feature in Amazon Web Services (AWS) that adds an extra layer of protection to an account. In addition to the account password, the user must provide a time-based verification code generated by an authenticator application or another supported MFA device.

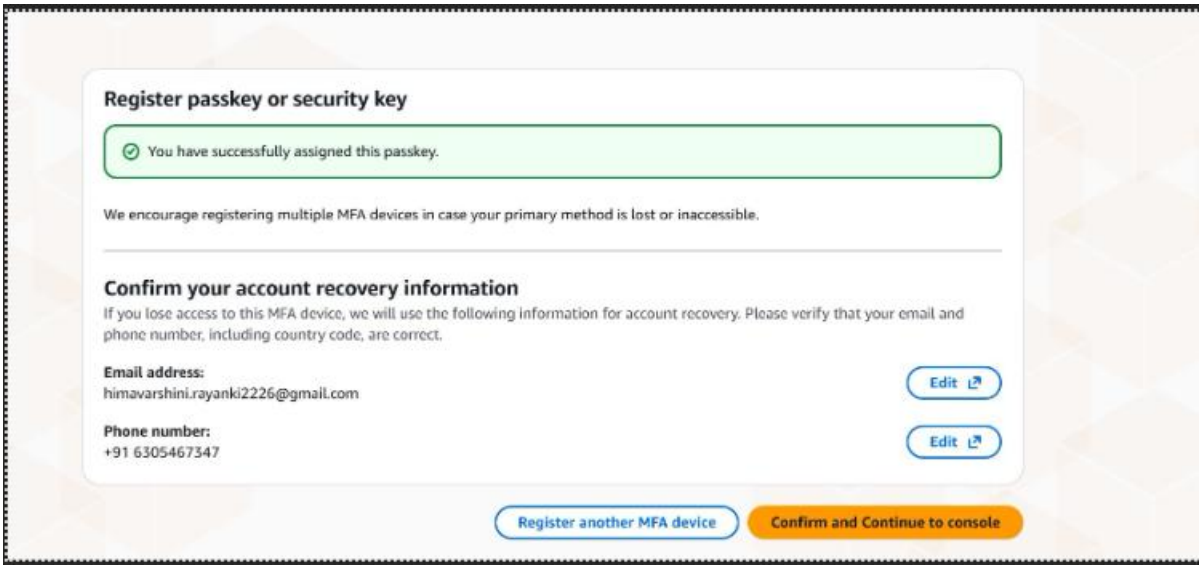
Enabling MFA helps protect the AWS account from unauthorized access, even if the password is known to someone else. It is recommended to enable MFA for the root user and other users who have access to important AWS resources.

Steps:

1. Sign in to the AWS Management Console using your account credentials.
2. Open the account menu and select Security credentials. For an IAM user, open the IAM service and select the required user.
3. Find the Multi-Factor Authentication (MFA) section and choose Assign MFA device.
4. Enter a name for the device, select an available MFA method, and continue.
5. If using an authenticator application, scan the displayed QR code with the application.
6. Enter the verification codes requested by AWS and select Add MFA.
7. Confirm that the MFA device is shown as assigned or enabled for the account.


aws
Root user sign in
Enter the password for
himavarshini.rayanki2226@gmail.com (not you?)
Password
Show password
Forgot password?
Sign in
Sign in to a different account
Create a new AWS account
AWS Builder Center
Skip setup, start building.
Free sandbox environments.
No credit card required.
Build up to 8 hours.
Explore Workshops →


MFA device name
Device Name
Maximum 128 characters. Use alphanumeric and '+ = , . @ - _' characters.
Passkey or Security key
Authenticate using your fingerprint, face, or screen lock. Create a passkey on this device or use another device, like a FIDO2 security key.
AWS recommends using a passkey or security key as an MFA device as they provide the most protection against attacks such as phishing.
Authenticator app
Authenticate using a code generated by an app installed on your mobile device or computer.
Hardware TOTP Token
Authenticate using a code generated by hardware TOTP token or other hardware devices.
Skip for now
Next


Register passkey or security key
You have successfully assigned this passkey.
We encourage registering multiple MFA devices in case your primary method is lost or inaccessible.
Confirm your account recovery information
If you lose access to this MFA device, we will use the following information for account recovery. Please verify that your email and phone number, including country code, are correct.
Email address:
himavarshini.rayanki2226@gmail.com
Edit
Phone number:
+91 6305467347
Edit
Register another MFA device
Confirm and Continue to console

1. Sign in to the AWS Management Console and open the Amazon EC2 service.
2. Select Launch instance and enter a suitable name for the Linux instance.
3. Choose a Linux Amazon Machine Image (AMI), such as Amazon Linux, and select an instance type.
4. Create a new key pair or select an existing key pair. Download and safely store the private key file if a new key pair is created.
5. Configure the network settings and security group. Allow SSH access on port 22 only from a trusted IP address.
6. Review the instance settings and choose Launch instance.
7. Open the Instances page and wait until the instance is running and its status checks are complete.
8. Select the instance and choose Connect. Follow the SSH connection instructions using the downloaded private key.

Output:

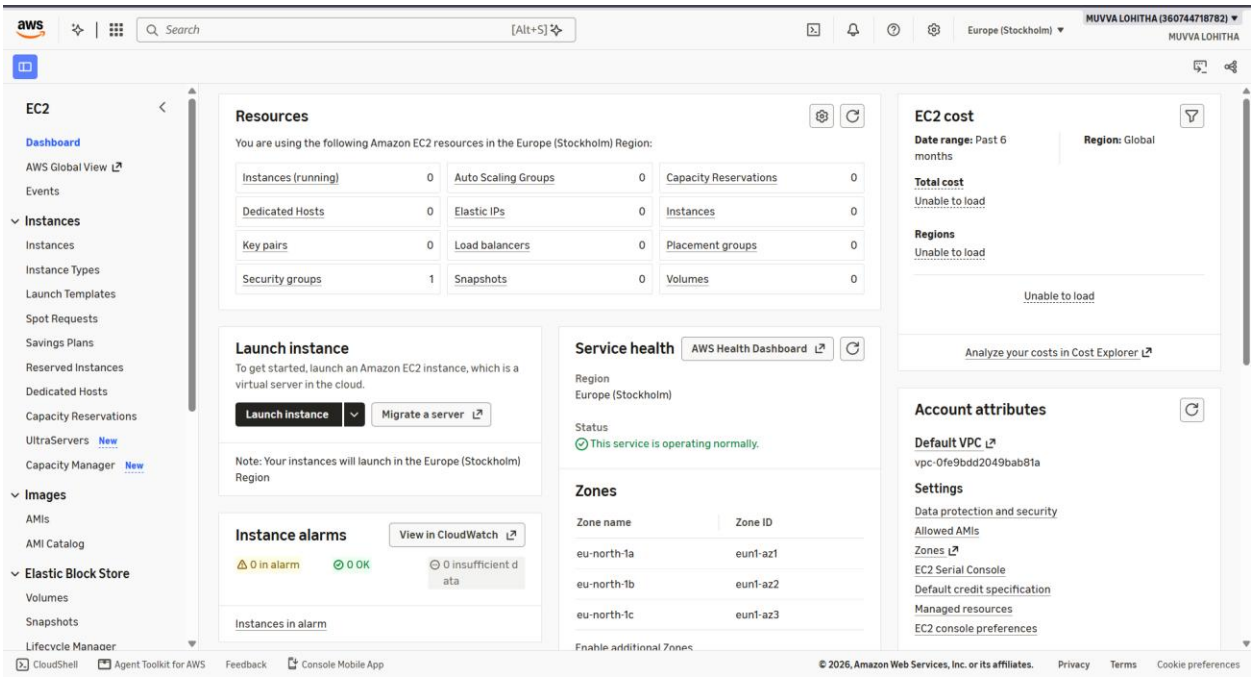
Output:MFA was successfully configured for the AWS account. The assigned MFA device is displayed in the AWS security settings, providing an additional verification step during sign-in.

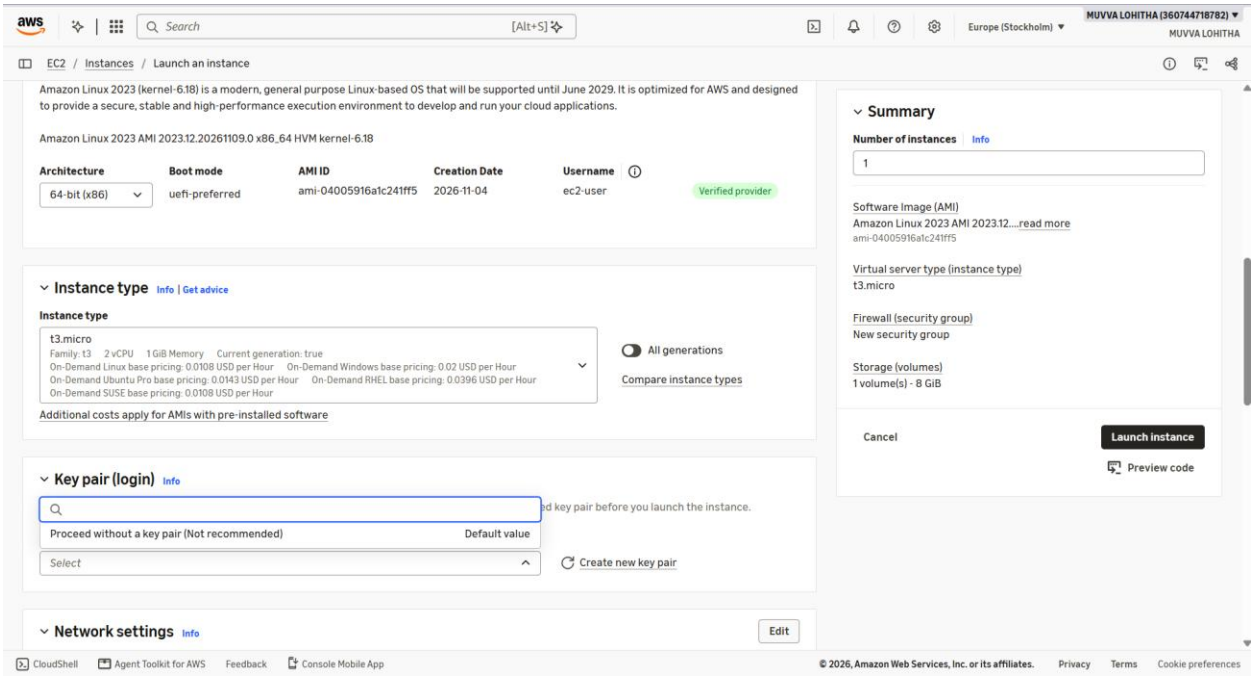
AIM: 2B : Creating First Linux Instance

Description:

Amazon Elastic Compute Cloud (Amazon EC2) is an AWS service used to create and manage virtual servers in the cloud. An EC2 Linux instance is a virtual machine that runs a Linux operating system and provides computing resources such as processor power, memory and storage.

To create an instance, select a Linux Amazon Machine Image (AMI), choose an instance type, configure storage and create or select a key pair. Configure the security group to allow the required network access, then launch the instance. After launch, connect to it securely using SSH and the private key associated with the key pair.


Search
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Europe (Stockholm)
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EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
UltraServers New
Capacity Manager New
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
Resources
You are using the following Amazon EC2 resources in the Europe (Stockholm) Region:
Instances (running) 0
Auto Scaling Groups 0
Capacity Reservations 0
Dedicated Hosts 0
Elastic IPs 0
Instances 0
Key pairs 0
Load balancers 0
Placement groups 0
Security groups 1
Snapshots 0
Volumes 0
EC2 cost
Date range: Past 6 months
Region: Global
Total cost
Unable to load
Regions
Unable to load
Unable to load
Analyze your costs in Cost Explorer
Launch instance
To get started, launch an Amazon EC2 instance, which is a virtual server in the cloud.
Launch instance
Migrate a server
Note: Your instances will launch in the Europe (Stockholm) Region
Service health
AWS Health Dashboard
Region
Europe (Stockholm)
Status
This service is operating normally.
Zones
Zone name
Zone ID
eu-north-1a
eun1-az1
eu-north-1b
eun1-az2
eu-north-1c
eun1-az3
Enable additional Zones
Account attributes
Default VPC
vpc-0fe9bdd2049bab81a
Settings
Data protection and security
Allowed AMIs
Zones
EC2 Serial Console
Default credit specification
Managed resources
EC2 console preferences
Instance alarms
View in CloudWatch
0 in alarm
0 OK
0 insufficient data
Instances in alarm
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
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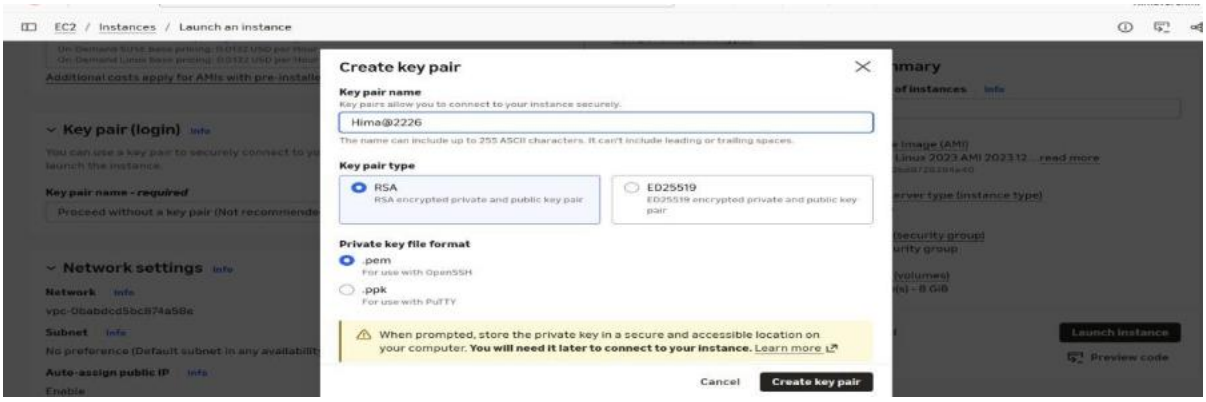

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EC2 / Instances / Launch an instance
Amazon Linux 2023 (kernel-6.18) is a modern, general purpose Linux-based OS that will be supported until June 2029. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.
Amazon Linux 2023 AMI 2023.12.20261109.0 x86_64 HVM kernel-6.18
Architecture
64-bit (x86)
Boot mode
uefi-preferred
AMI ID
ami-04005916a1c241ff5
Creation Date
2026-11-04
Username
ec2-user
Verified provider
Instance type
Info | Get advice
Instance type
t3.micro
Family: t3 2 vCPU 1 GiB Memory Current generation: true
On-Demand Linux base pricing: 0.0108 USD per Hour On-Demand Windows base pricing: 0.02 USD per Hour
On-Demand Ubuntu Pro base pricing: 0.0143 USD per Hour On-Demand RHEL base pricing: 0.0396 USD per Hour
On-Demand SUSE base pricing: 0.0108 USD per Hour
All generations
Compare instance types
Additional costs apply for AMIs with pre-installed software
Key pair (login)
Info
Proceed without a key pair (Not recommended)
Default value
Select
Create new key pair
Network settings
Info
Edit
Summary
Number of instances
Info
1
Software Image (AMI)
Amazon Linux 2023 AMI 2023.12....read more
ami-04005916a1c241ff5
Virtual server type (instance type)
t3.micro
Firewall (security group)
New security group
Storage (volumes)
1 volume(s) - 8 GiB
Cancel
Launch instance
Preview code
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
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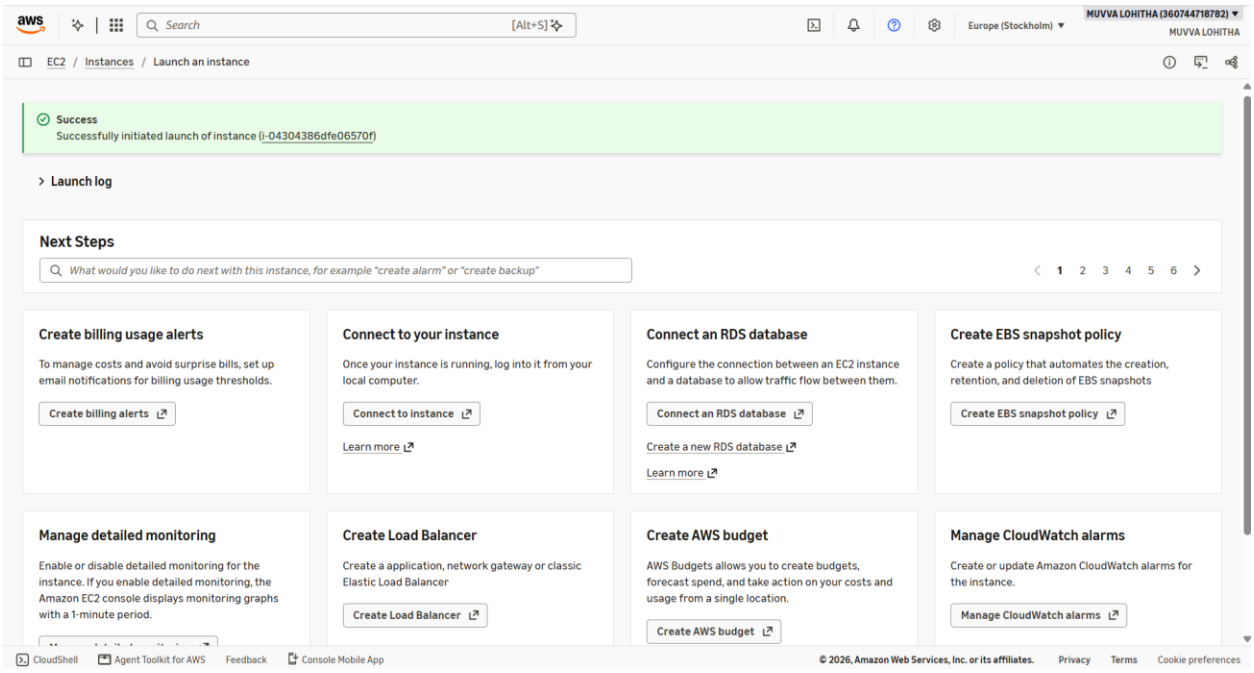
Steps:

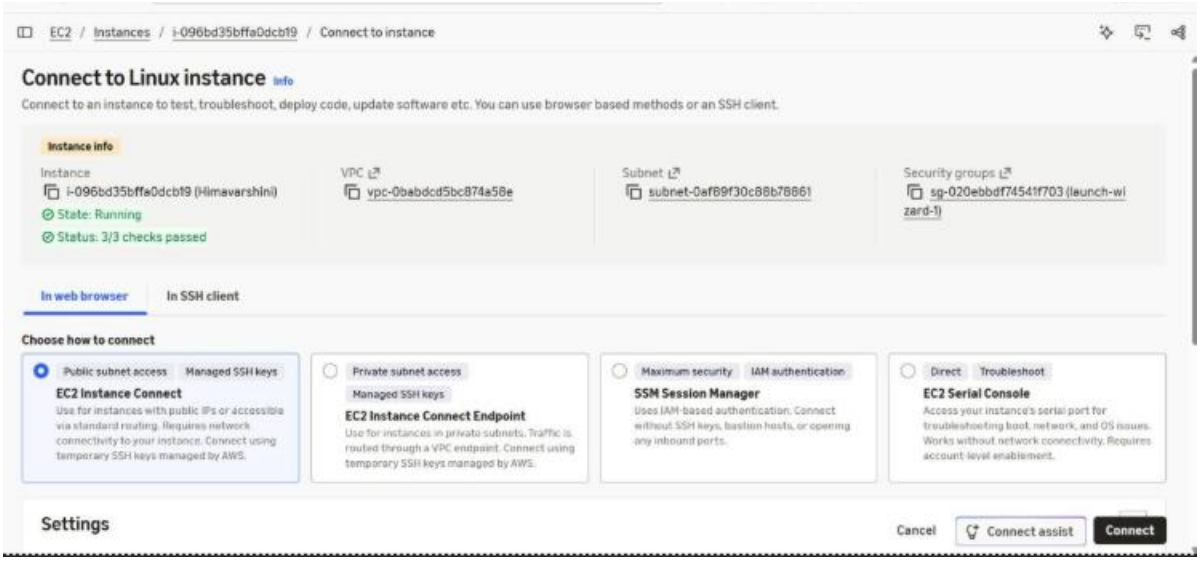
1. Sign in to the AWS Management Console and open the Amazon S3 service.
2. Select Create bucket.
3. Enter a globally unique bucket name and choose the required AWS Region.
4. Keep the recommended security settings, including Block all public access, unless the lab instructions require otherwise.
5. Review the settings and select Create bucket.
6. Open the newly created bucket and select Upload.
7. Choose Add files and select five files with different extensions, for example PDF, JPG, PNG, TXT and DOCX.
8. Review the selected files and choose Upload.
9. Wait for the upload to complete, then verify that all five objects are listed in the bucket.

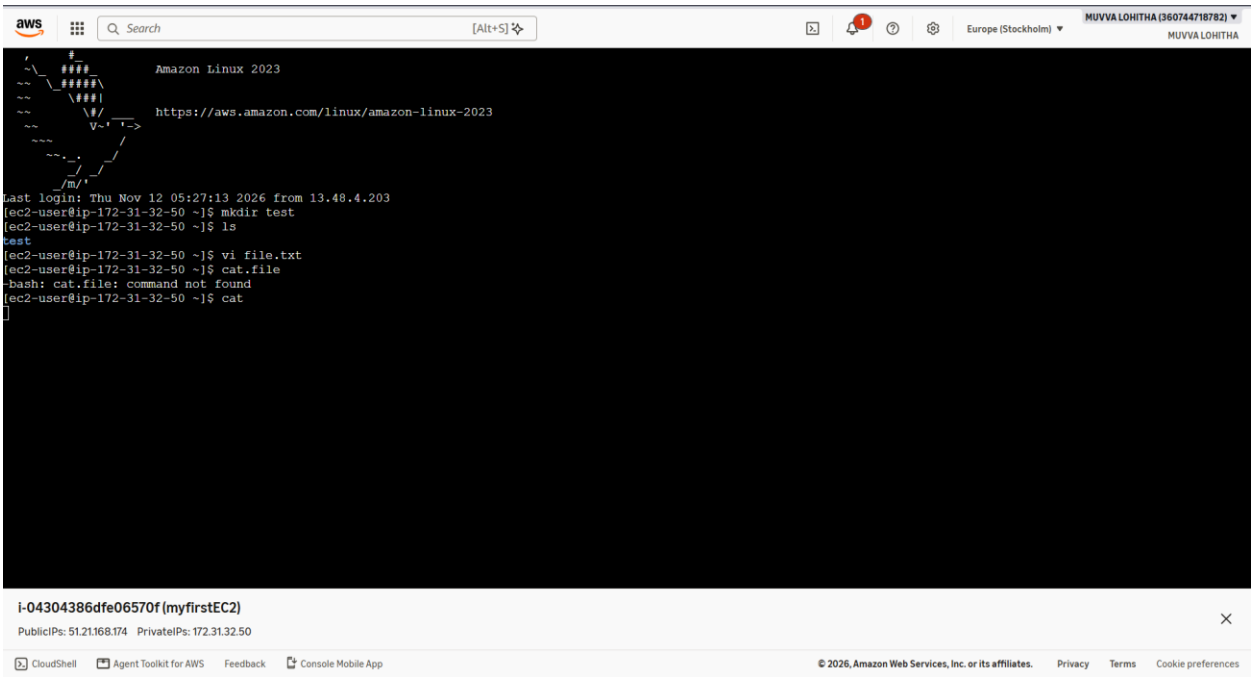
Output:

Output:The Linux EC2 instance was successfully launched. Its instance ID, instance type, and current state are displayed on the EC2 Instances page. The instance state should show “Running” after the launch is complete.




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EC2 / Instances / Launch an instance
Success
Successfully initiated launch of instance (i-04304386dfe06570f)
Launch log
Next Steps
What would you like to do next with this instance, for example "create alarm" or "create backup"
1 2 3 4 5 6
Create billing usage alerts
To manage costs and avoid surprise bills, set up email notifications for billing usage thresholds.
Create billing alerts
Connect to your instance
Once your instance is running, log into it from your local computer.
Connect to instance
Learn more
Connect an RDS database
Configure the connection between an EC2 instance and a database to allow traffic flow between them.
Connect an RDS database
Create a new RDS database
Learn more
Create EBS snapshot policy
Create a policy that automates the creation, retention, and deletion of EBS snapshots
Create EBS snapshot policy
Manage detailed monitoring
Enable or disable detailed monitoring for the instance. If you enable detailed monitoring, the Amazon EC2 console displays monitoring graphs with a 1-minute period.
Create Load Balancer
Create a application, network gateway or classic Elastic Load Balancer
Create Load Balancer
Create AWS budget
AWS Budgets allows you to create budgets, forecast spend, and take action on your costs and usage from a single location.
Create AWS budget
Manage CloudWatch alarms
Create or update Amazon CloudWatch alarms for the instance.
Manage CloudWatch alarms
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
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EC2 / Instances / i-096bd35bffa0dcb19 / Connect to instance
Connect to Linux instance Info
Connect to an instance to test, troubleshoot, deploy code, update software etc. You can use browser based methods or an SSH client.
Instance info
Instance
i-096bd35bffa0dcb19 (Himavarshini)
State: Running
Status: 3/3 checks passed
VPC
vpc-0babdcd5bc874a58e
Subnet
subnet-0af69f30c86b78661
Security groups
sg-020ebbdf74541f703 (launch-wizard-1)
In web browser
In SSH client
Choose how to connect
Public subnet access
Managed SSH keys
EC2 Instance Connect
Use for instances with public IPs or accessible via standard routing. Requires network connectivity to your instance. Connect using temporary SSH keys managed by AWS.
Private subnet access
Managed SSH keys
EC2 Instance Connect Endpoint
Use for instances in private subnets. Traffic is routed through a VPC endpoint. Connect using temporary SSH keys managed by AWS.
Maximum security
IAM authentication
SSM Session Manager
Uses IAM-based authentication. Connect without SSH keys, bastion hosts, or opening any inbound ports.
Direct
Troubleshoot
EC2 Serial Console
Access your instance's serial port for troubleshooting boot, network, and OS issues. Works without network connectivity. Requires account-level enablement.
Settings
Cancel
Connect assist
Connect



Output:

Output:The S3 bucket was successfully created, and five objects with different file extensions were uploaded. All five files are displayed in the bucket's Objects list and are available for further management.

AIM: Bucket Creation and Upload Objects

Description:

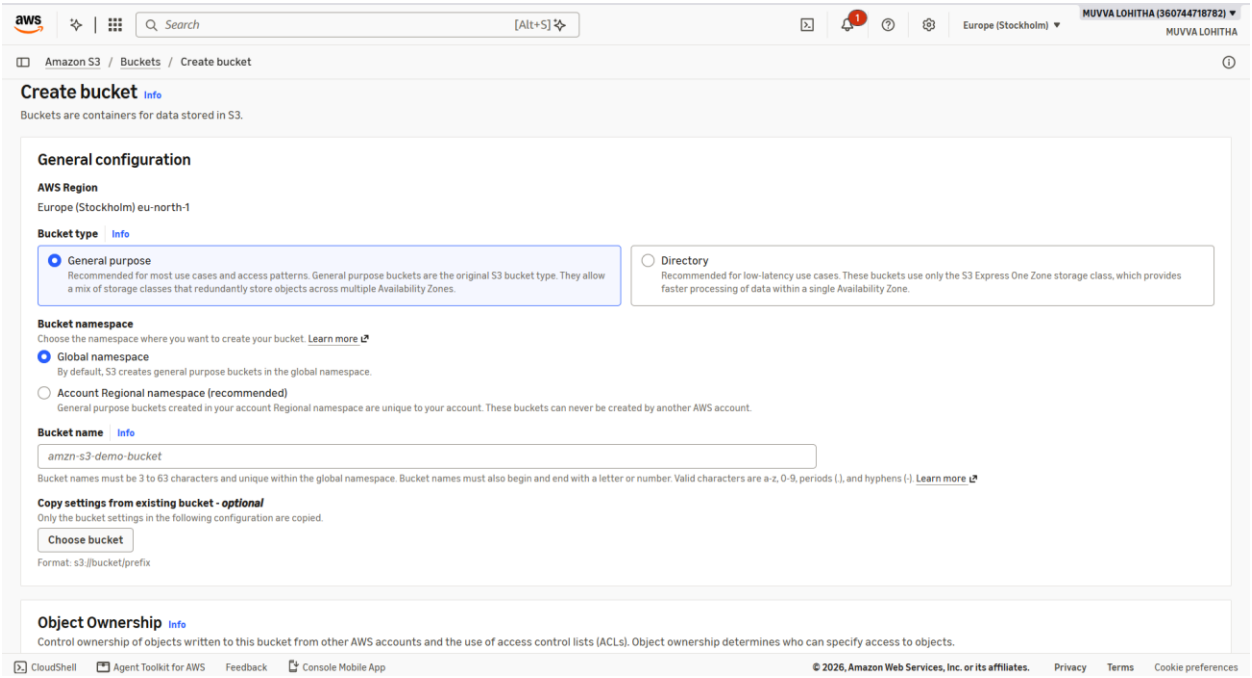
Amazon Simple Storage Service (Amazon S3) is an object storage service provided by AWS. It allows users to store and manage files in containers called buckets. A bucket is created with a globally unique name and an AWS Region, and it can store documents, images, videos and other types of files.

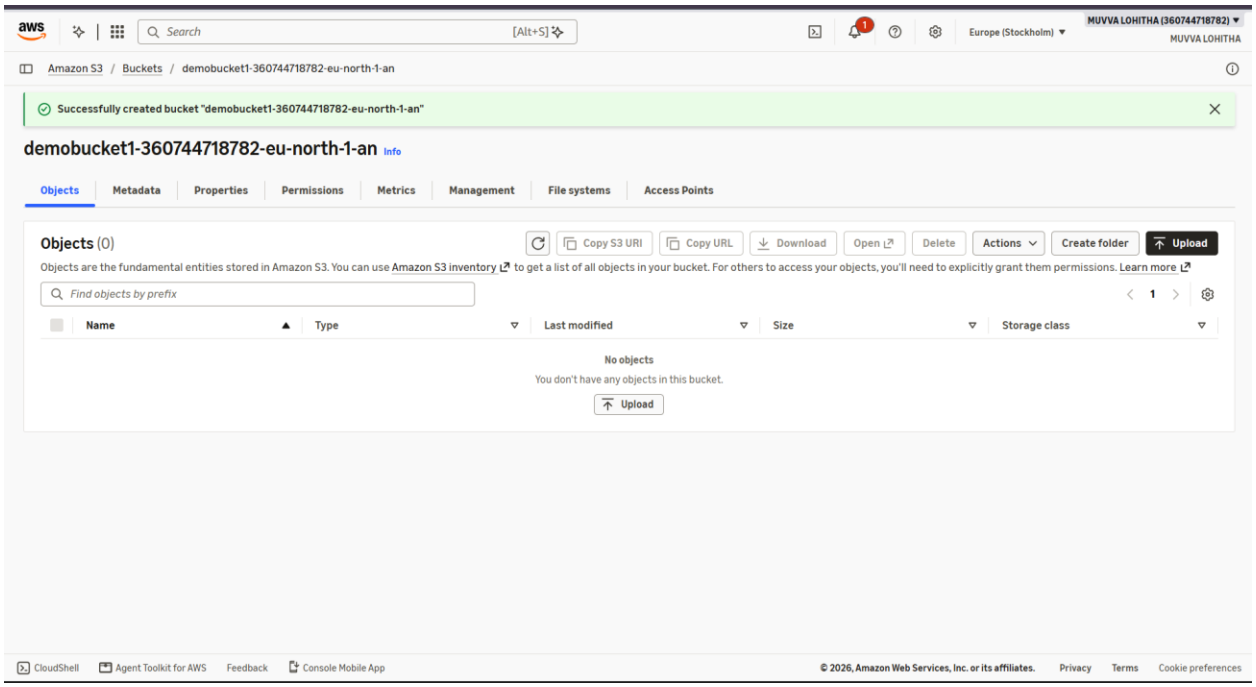
After creating the bucket, users can upload objects in different file formats. In this experiment, five files with different extensions, such as PDF, JPG, PNG, TXT and DOCX, are uploaded. The uploaded objects can be viewed and managed from the S3 bucket.

Steps:

Sign in to the AWS Management Console and open the Amazon S3 service.

2. Click Create Bucket and enter a unique bucket name.
3. Select the AWS Region and keep the default settings as required.
4. Click Create Bucket to create the storage bucket.
5. Open the created bucket and click Upload.
6. Click Add Files and select five files with different extensions, such as .pdf, .jpg, .txt, .docx, and .png.
7. Click Upload and wait until all five objects are successfully uploaded.





Output:

The Amazon S3 bucket was successfully created for cloud file storage. Five objects with different file extensions were uploaded successfully and stored in the bucket.